

Michael Smith
SedLab
5/3/2026

1. Sed 's/Jo/Josephine/g'

This command replaced all instances of Jo with Josephine. The s tells it to substitute the old word /Jo with the new word /Josephine, and the g tells it to substitute all instances, not just once.

<https://www.geeksforgeeks.org/linux-unix/sed-command-in-linux-unix-with-examples/>

```
ms3213@UbuntuServer1:~$ sed 's/Jo/Josephine/g' SedLab_
Josephine March:327-832-5728:3465 Mirlo Street, Peabody, MA 34756:10/2/65:35200
Victor Frankenstein:835-365-1284:454 Easy Street, Decatur, IL 75732:2/28/53:123500
```

2.sed '\$d' SedLab.txt | sed '\$d' | sed '\$d' | sed '\$d' | sed '\$d'

In this command, \$ stands for the last line, and d stands for delete of the SedLab file. Using | we use piping to give the output of the command we just ran to the next 'sed \$d'. This basically makes a chain where, after the last line is deleted, the output is passed along, so the next-to-last line is now the last line. This sed '\$d' repeats 4 times after the first to successfully delete 5 lines off the end of sedlab

<https://www.geeksforgeeks.org/linux-unix/sed-command-in-linux-unix-with-examples/>

<https://stackoverflow.com/questions/13380607/how-to-use-sed-to-remove-the-last-n-lines-of-a-file>

<https://www.geeksforgeeks.org/linux-unix/piping-in-unix-or-linux/>

```
> sed '$d' SedLab.txt | sed '$d' | sed '$d' | sed '$d' | sed '$d'
Huckleberry Finn:238-923-7366:95 Latham Lane, Easton, PA 83755:11/12/56:20300
Betty Boop:245-836-8357:635 Cutesy Lane, Hollywood, CA 91464:6/23/23:14500
Dorian Gray:385-375-8395:3567 Populus Place, Caldwell, NJ 23875:6/18/68:23400
Holly Golightly:397-857-2735:74 Pine Street, Dearborn, MI 23874:3/28/45:245700
Ebenezer Scrooge:548-834-2348:583 Laurel Ave., Kingsville, TX 83745:10/1/35:58900
Hester Prynne:408-253-3122:123 Park St., San Jose, CA 04086:7/25/53:85100
Westley Pirate:284-758-2857:23 Edgecliff Place, Lincoln, NB 92743:7/25/53:85100
Westley Pirate:284-758-2867:23 Edgecliff Place, Lincoln, NB 92743:11/3/35:58200
Westley Pirate:284-758-2867:23 Edgecliff Place, Lincoln, NB 92743:11/3/35:58200
Lizzie Bennett:674-843-1385:20 Parak Lane, Duluth, MN 23850:4/12/23:780900
Nancy Drew:674-843-1385:20 Parak Lane, Duluth, MN 23850:4/12/23:780900
Jo March:327-832-5728:3465 Mirlo Street, Peabody, MA 34756:10/2/65:35200
Victor Frankenstein:835-365-1284:454 Easy Street, Decatur, IL 75732:2/28/53:123500
Ephram Hardy:293-259-5395:235 Carlton Lane, Joliet, IL 73858:8/12/20:56700
Meg Murry:834-938-8376:23445 Aster Ave., Allentown, NJ 83745:12/1/38:45000
Lucy Pevensie:385-573-8326:832 Ponce Drive, Gzary, IN 83756:12/1/46:268500
Hemione Granger:408-456-1234:4 Harvard Square, Boston, MA 02133:4/22/62:52600
Mary Poppins:846-836-2837:6937 Ware Road, Milton, PA 93756:9/21/46:43500
Sir Lancelot:837-835-8257:474 Camelot Boulevard, Bath, WY 28356:5/13/69:24500
Minerva McGonagall:408-233-8971:45 Rose Terrace, San Francisco, CA 92303:2/3/36:25000
Zippy Pinhead:834-823-8319:2356 Bizarro Ave., Farmount, IL 84357:1/1/67:89500
Dorothy Gale:923-835-8745:23 Wimp Lane, Kensington, DL 38758:8/31/69:126000
Popeye Sailor:156-454-3322:945 Bluto Street, NoHere, USA 29358:3/19/35:22350
Luna Lovegood:385-898-8357:38 Fife Way, Abilene, TX 39673:1/5/58:95600
```

3. sed -n '3,15p' SedLab

<https://how.dev/answers/what-is-the-sed-pattern-range>

The -n in this command makes sure it only prints everything in the command asked for, as without it would print all the lines. The '3,15 specifies lines 3-15 and the p tells it to print.

```
ms3213@UbuntuServer1:~$ sed -n '3,15p' SedLab
Betty Boop:245-836-8357:635 Cutesy Lane, Hollywood, CA 91464:6/23/23:14500
Dorian Gray:385-375-8395:9567 Populus Place, Caldwell, NJ 23875:6/18/68:23
Holly Golightly:397-857-2735:74 Pine Street, Dearborn, MI 23874:3/28/45:24
Ebenezer Scrooge:548-834-2348:583 Laurel Ave., Kingsville, TX 83745:10/1/3
Hester Prynne:408-253-3122:123 Park St., San Jose, CA 04086:7/25/53:85100
Westley Pirate:284-758-2857:23 Edgecliff Place, Lincoln , NB 92743:7/25/53
Westley Pirate:284-758-2867:23 Edgecliff Place, Lincoln, NB 92743:11/3/35:
ms3213@UbuntuServer1:~$ _
```

4. sed '/CA/d' SedLab

The '/CA/' specifies that it is looking for lines with CA in it, which stands for California. The d' tells it to delete these lines that contain California. Pictured shows it printing the lines without the lines that contain CA.

<https://stackoverflow.com/questions/5410757/how-to-delete-from-a-text-file-all-lines-that-contain-a-specific-string>

```
ephram Hardy:293-259-5395:235 Carlton Lane, Joliet, IL 73858:8/12/20:56700
Meg Murry:834-938-8376:23445 Aster Ave., Allentown, NJ 83745:12/1/38:45000
Lucy Pevensie:385-573-8326:832 Ponce Drive, Gzary, IN 83756:12/1/46:268500
Remione Granger:408-456-1234:4 Harvard Square, Boston, MA 02133:4/22/62:52600
Mary Poppins:846-836-2837:6937 Ware Road, Milton, PA 93756:9/21/46:43500
Sir Lancelot:837-835-8257:474 Camelot Boulevard, Bath, WY 28356:5/13/69:24500
Minerva McGonagall:408-233-8971:45 Rose Terrace, San Francisco, CA 92303:2/3/36
Dippy Pinhead:834-823-8319:2356 Bizarro Ave., Farmount, IL 84357:1/1/67:89500
Dorothy Gale:923-835-8745:23 Wimp Lane, Kensington, DL 38758:8/31/69:126000
Popeye Sailor:156-454-3322:945 Bluto Street, NotHere, USA 29358:3/19/35:22350
Luna Lovegood:385-898-8357:38 Fife Way, Abilene, TX 39673:1/5/58:95600
Daenerys Targaryen:408-724-0140:1222 Oxbow Court, Sunnyvale, CA 94087:5/19/66:3
Molly Weasley:387-827-1095:13 Uno Lane, Ashville, NC 23556:7/1/29:57000
Arya Stark:438-910-7449:8235 Maple Street, Wilmington, VM 29085:9/23/63:68900
ms3213@UbuntuServer1:~$ sed '/California/d' SedLab
```

5. Sed -r -n '/([0-9]){2})V[1-7]V/p' SedLab

The -r allows use of extended regex so that we can use the {2} to repeat a number pattern. The -n cancels the automatic printing, so it only prints what we ask. In quotes, we have [0-9], which means any number 0-9. Then | which means 'or' followed by [0-9]{2}, which means any 2 digit number. The {2} repeats the [0-9]. We then have to use the delimiter \ so that '/' is taken as part

of the pattern. Now [1-7] is used to look for just the first 7 days of the month. We then have a delimiter \ then / because it is part of the date pattern. The / at the end closes the pattern and is not part of the pattern. The p at the end tells it to print this pattern of the file SedLab.

https://developer.mozilla.org/en-US/docs/Web/JavaScript/Guide/Regular_expressions/Cheatsheet
https://www.gnu.org/software/sed/manual/sed.html#Command_002dLine-Options

```
ms3213@UbuntuServer1:~$ sed -r -n '/([0-9]|[0-9]{2})\[1-7]\[/p' SedLab
Ebenezer Scrooge:548-834-2348:583 Laurel Ave., Kingsville, TX 83745:10/1/35:58900
Westley Pirate:284-758-2867:23 Edgecliff Place, Lincoln, NB 92743:11/3/35:58200
Westley Pirate:284-758-2867:23 Edgecliff Place, Lincoln, NB 92743:11/3/35:58200
Jo March:327-832-5728:3465 Mirlo Street, Peabody, MA 34756:10/2/65:35200
Meg Murry:834-938-8376:23445 Aster Ave., Allentown, NJ 83745:12/1/38:45000
Lucy Pevensie:385-573-8326:832 Ponce Drive, Gzary, IN 83756:12/1/46:268500
Minerva McGonagall:408-233-8971:45 Rose Terrace, San Francisco, CA 92303:2/3/36:
Zippy Pinhead:834-823-8319:2356 Bizarro Ave., Farmount, IL 84357:1/1/67:89500
Luna Lovegood:385-898-8357:38 Fife Way, Abilene, TX 39673:1/5/58:95600
Molly Weasley:387-827-1095:13 Uno Lane, Ashville, NC 23556:7/1/29:57000
ms3213@UbuntuServer1:~$
```

6. sed '/^Sir/s/\$/****' SedLab

The command starts with the carrot to tell it to look at the beginning of the lines for Sir. The s command tells it to substitute/add something to these lines, and the \$ tells it to put it at the end. Finally, the **** is what is being added at the end.

<https://askubuntu.com/questions/537967/append-to-end-of-a-line-using-sed>

```
ms3213@UbuntuServer1:~$ sed '/^Sir/s/$/****/' SedLab_
```

7. sed '/Westley Pirate/c\As you wish/' SedLab

This command says to look for all instances of Westley Pirate with /Westley Pirate/ then c\ tells it to replace the whole line. Putting As you wish/ after tells it to replace the whole line with “As you wish”.

<https://stackoverflow.com/questions/11245144/replace-whole-line-containing-a-string-using-sed>

```
Hester Prynne:408-253-8122:123 Park St., San Jose, CA 04086:7/25/53:85100
As you wish/
As you wish/
As you wish/
Lizzie Bennett:674-843-1385:20 Parak Lane, Duluth, MN 23850:4/12/23:780900
```

8. **sed -r '/Minerva McGonagall/s/([0-9][0-9]{2}\([0-9][0-9]{2}\V[0-9]{2}\)/10V04V1935/'** SedLab

The -r allows the command to use repetition in the regex parts {2}. We first start with Minerva McGonagall's name to look for that specific line. Then the /s/ tells it that something in the line is going to be substituted. Next ([0-9][0-9]{2}\([0-9][0-9]{2}\V[0-9]{2}) is the pattern for searching for the birthday as it is in the format mm/dd/yy. ([0-9][0-9]{2} represents any number 0-9 or (|) any 2 numbers 0-9 for the month. Then the next ([0-9][0-9]{2} any number 0-9 or represented by | any 2 numbers 0-9 for the day. I use the | because many of the dates have a single digit for the days and months. The V is a delimiter with a / to allow for the / to be part of the pattern. Then the last [0-9]{2} represents any 2 numbers 0-9 for the year. Right after that /10V04V1935/' is used to represent the new date of 10/04/1935 that we want to substitute into that pattern.

<https://askubuntu.com/questions/537967/appending-to-end-of-a-line-using-sed>

```
ms3213@UbuntuServer1:~$ sed -r '/Minerva McGonagall/s/([0-9][0-9]{2}\([0-9][0-9]{2}\V[0-9]{2}\)/10V04V1935/' SedLab_
```

```
Minerva McGonagall:408-233-8971:45 Rose Terrace, San Francisco, CA 92303:10/04/1935:25000
```

9.

sed '/^\$/d' SedLab

This command starts with the ^, which means start at the beginning of the line, then is followed by \$, which means the end of the line. Because there is nothing in between these two symbols, it matches any empty lines. The /d' afterwards says to delete these empty lines.

```
msmith@UbuntuServer1:~$ sed '/^$/d' SedLab.txt
Huckleberry Finn:238-923-7366:95 Latham Lane, Easton, PA 83755:11/12/56:20300
Betty Boop:245-836-8357:635 Cutesy Lane, Hollywood, CA 91464:6/23/23:14500
Dorian Gray:385-375-8395:3567 Populus Place, Caldwell, NJ 23875:6/18/68:23400
Holly Golightly:397-857-2735:74 Pine Street, Dearborn, MI 23874:3/28/45:245700
Ebenezer Scrooge:548-834-2348:583 Laurel Ave., Kingsville, TX 83745:10/1/35:58900
Hester Prynne:408-253-3122:123 Park St., San Jose, CA 04086:7/25/53:85100
Westley Pirate:284-758-2857:23 Edgecliff Place, Lincoln , NB 92743:7/25/53:85100
Westley Pirate:284-758-2867:23 Edgecliff Place, Lincoln, NB 92743:11/3/35:58200
Westley Pirate:284-758-2867:23 Edgecliff Place, Lincoln, NB 92743:11/3/35:58200
Lizzie Bennett:674-843-1385:20 Parak Lane, Duluth, MN 23850:4/12/23:780900
Nancy Drew:674-843-1385:20 Parak Lane, Duluth, MN 23850:4/12/23:780900
Jo March:327-832-5728:3465 Mirlo Street, Peabody, MA 34756:10/2/65:35200
Victor Frankenstein:835-365-1284:454 Easy Street, Decatur, IL 75732:2/28/53:123500
Ephram Hardy:293-259-5395:235 Carlton Lane, Joliet, IL 73858:8/12/20:56700
Meg Murry:834-938-8376:23445 Aster Ave., Allentown, NJ 83745:12/1/38:45000
Lucy Pevensie:385-573-8326:832 Ponce Drive, Gzary, IN 83756:12/1/46:268500
Hemione Granger:408-456-1234:4 Harvard Square, Boston, MA 02133:4/22/62:52600
Mary Poppins:846-836-2837:6937 Ware Road, Milton, PA 93756:9/21/46:43500
Sir Lancelot:837-835-8257:474 Camelot Boulevard, Bath, WY 28356:5/13/69:24500
Minerva McGonagall:408-233-8971:45 Rose Terrace, San Francisco, CA 92303:2/3/36:25000
Zippy Pinhead:834-823-8319:2356 Bizarro Ave., Farmount, IL 84357:1/1/67:89500
Dorothy Gale:923-835-8745:23 Wimp Lane, Kensington, DL 38758:8/31/69:126000
Popeye Sailor:156-454-3322:945 Bluto Street, NotHere, USA 29358:3/19/35:22350
Luna Lovegood:385-898-8357:38 Fife Way, Abilene, TX 39673:1/5/58:95600
Daenerys Targaryen:408-724-0140:1222 Oxbow Court, Sunnyvale, CA 94087:5/19/66:34200
Molly Weasley:387-827-1095:13 Uno Lane, Ashville, NC 23556:7/1/29:57000
Arya Stark:438-910-7449:8235 Maple Street, Wilmington, VM 29085:9/23/63:68900
```

10.

1i\ Great Literary Characters -

The 1 in this command tells the location as the first line, and the i says to insert something before this first line. Followed by the Great Literary Characters, which is what is inserted

s/[0-9]{3}-[0-9]{3}-[0-9]{4}/1+&/

This command starts with s, which allows for substitution. Followed by the pattern that represents phone numbers. [0-9]{3}- is Any digit 0-9 for three times and a dash '-' repeat that again for another 3. Then any 4 digits is [0-9]{4}/. The 1+ is the text we are adding, and the & tells it to add it to the beginning of the said pattern.

\$ a Happily Ever after. The End

The \$ symbol refers to the last line of the file, and the 'a' tells it to append something after this line. The Happily ever after is what is being appended.

```

- Great Literary Characters -
Huckleberry Finn:1+238-923-7366:95 Latham Lane, Easton, PA 83755:11/12/56:20300

Betty Boop:1+245-836-8357:635 Cutesy Lane, Hollywood, CA 91464:6/23/23:14500

Dorian Gray:1+385-375-8395:3567 Populus Place, Caldwell, NJ 23875:6/18/68:23400

Holly Golightly:1+397-857-2735:74 Pine Street, Dearborn, MI 23874:3/28/45:245700

Ebenezer Scrooge:1+548-834-2348:583 Laurel Ave., Kingsville, TX 83745:10/1/35:58900

Hester Prynne:1+408-253-3122:123 Park St., San Jose, CA 04086:7/25/53:85100

Westley Pirate:1+284-758-2857:23 Edgecliff Place, Lincoln , NB 92743:7/25/53:85100

Westley Pirate:1+284-758-2867:23 Edgecliff Place, Lincoln, NB 92743:11/3/35:58200

Marg Poppins:1+848-838-2857:8937 Rare Road, Milton, TN 93758:9/21/48:49300

Sir Lancelot:1+837-835-8257:474 Camelot Boulevard, Bath, WY 28356:5/13/69:24500

Minerva McGonagall:1+408-233-8971:45 Rose Terrace, San Francisco, CA 92303:2/3/36:0000

Zippy Pinhead:1+834-823-8319:2356 Bizarro Ave., Farmount, IL 84357:1/1/67:89500

Dorothy Gale:1+923-835-8745:23 Wimp Lane, Kensington, DL 38758:8/31/69:126000

Popeye Sailor:1+156-454-3322:945 Bluto Street, NoWhere, USA 29358:3/19/35:22350

Luna Lovegood:1+385-898-8357:38 Fife Way, Abilene, TX 39673:1/5/58:95600

Daenerys Targaryen:1+408-724-0140:1222 Oxbow Court, Sunnyvale, CA 94087:5/19/66:34000

Molly Weasley:1+387-827-1095:13 Uno Lane, Ashville, NC 23556:7/1/29:57000

Arya Stark:1+438-910-7449:8235 Maple Street, Wilmington, VM 29085:9/23/63:68900

Happily Ever after. The End

```

```

1 | li\ - Great Literary Characters -
2 | s/[0-9]{3}-[0-9]{3}-[0-9]{4}/1+&/
3 | $ a Happily Ever after. The End

```

Work revised 5/3

<https://unix.stackexchange.com/questions/99350/how-to-insert-text-before-the-first-line-of-a-file>
<https://unix.stackexchange.com/questions/121161/how-to-insert-text-after-a-certain-string-in-a-file>
https://www.gnu.org/software/sed/manual/sed.html#The-_0022s_0022-Command
<https://unix.stackexchange.com/questions/20573/sed-insert-text-after-the-last-line>